

Ali Mamaghani | Curriculum Vitae

☎ (+1) 858 829 2270 • ✉ amamaghani@ucsd.com • in ali-mamaghani

Education

University of California San Diego <i>Ph.D. in Electrical and Computer Engineering</i> GPA-3.9/4	San Diego-USA 2024-Present
Sharif University of Technology <i>Masters in Electrical Engineering (Communication Systems)</i> GPA-19.11/20.00(4/4)	Tehran-Iran 2022-Present
Sharif University of Technology <i>Bachelors in Electrical Engineering (Communication Systems)</i> GPA-19.15/20.00(3.99/4)	Tehran-Iran 2018-2022
Allameh Helli High School <i>Mathematics and Physics Diploma.</i>	Tehran-Iran 2014-2018
Young Scholars Club <i>Physics Olympiad Courses</i>	Tehran-Iran Jun 2017- Sep 2017

Research Interests

- Communication Networks
- Machine Learning
- Optimization
- Reinforcement Learning
- Information Theory
- SDN/NFV
- Queuing Theory
- Game Theory
- AR/VR

Research Experience

Question Answering over Telecom Log Systems <i>Research Project</i>	<i>Jan 2025 - Present</i>
○ <i>Supervisor:</i> Dinesh Bharadia ○ Proposed PRISM, a two-stage framework combining causal-semantic log pruning and LLM-based reasoning for interpretable 5G system analysis. ■ Designed a log-pruning pipeline that uses a RAG approach to condense millions of log lines into query-aligned templates, delivering up to 30% higher factual accuracy and 66% lower compute time. ■ Developed a hypothesis-verification reasoning algorithm integrated with a 3GPP-derived knowledge graph, enabling log causal reasoning, diagnosing issues: buffer overflow, RACH failures, bad channels, HARQ retransmission, etc. ■ Validated the framework on over-the-air srsRAN testbeds and created a multi-scenario 5G log dataset covering both SDR and commercial RU deployments for reproducible evaluation.	
Edgeric for OAI <i>Research Project</i>	<i>Sep 2024 - Jan 2025</i>
○ <i>Supervisor:</i> Dinesh Bharadia ○ In this project I integrate Edgeric (A real-time radio intelligence controller) into the OAI 5G RAN, trying to increase user throughput using RL algorithms, and providing real-time monitoring for the fine-grain measurements of UE. This leads to application-aware optimization of RAN.	
Applications of LLMs in a Cloud-Native 5G platform form Eurecom <i>Master's Thesis Project</i>	<i>Apr 2023 - Aug 2024</i>
○ <i>Supervisors:</i> Prof. Hosseinkhalaj and Prof. Nikaein (from Eurecom) ○ In this project, first, I build a 5G network using Kubernetes and 5G components from OAI, then I provide AI power for different aspects of the network including providing explanations for the output of tests and errors, correcting configuration files using GPT3.5-turbo in a feedback loop and fine-tuning GPT3.5-turbo to generate configurations and develop a chatbot as an assistant.	
Implementation of 4G and 5G mobile Networks with centralized controller <i>Bachelor's Thesis Project</i>	<i>Agu 2021 - Jun 2022</i>
○ <i>Supervisor:</i> Prof. Hosseinkhalaj ○ In this project, I managed to implement a 4G network and different scenarios of 5G Networks(SA, NSA). Furthermore, I used O-ran RIC and slicing XApp in this network. In these implementations, I used OAI, Felex RAN/RIC from Mosaic5G, and Open5Gs platforms.	

Teaching Experience

- Probability and statistics**, Instructor: Prof. Ashtiani *Fall 2023*
○ Designed problem sets, held Q&A sessions, and graded homework of +20 students
- Data Networks**, Instructor: Prof. Pakraven *Fall 2022*
○ Designed problem sets, held Q&A sessions, and graded homework of +30 students
- Random Processes**, Instructor: Prof. Behnia *Spring 2022*
○ Held Q&A sessions, and graded homework of +20 students
- Data Networks**, Instructor: Prof. Pakraven *Spring 2022*
○ Designed problem sets, held Q&A sessions, and graded homework of +30 students
- Discrete-time Signal Processing**, Instructor: Prof. Shamsollahi *Fall 2021*
○ Designed problem sets and held Q&A sessions with +30 students
- Circuits Theory**, Instructor: Prof. Shamsollahi *Fall 2020*
○ Designed problem sets and held Q&A sessions with +30 students
- Electromagnetics**, Instructor: Prof. Memarian *Spring 2020*
○ Designed course website, designed problem sets, held Q&A sessions, and graded homework of +120 students

Honours and Awards

- UC San Diego Fellowship Winner *Feb 2024*
- **Ranked 2nd** among Electrical Engineering-communication systems master's Students *Nov 2023*
- **Ranked 5th** among 176 Electrical Engineering Students *2019 -2022*
- Nominated for Best BSc project awards. *Apr 2022*
- Direct admission for MSc from Sharif university of technology *Feb 2022*
- **Silver Medal** in the 31st National Physics Olympiad *2017*
This exam is annually held among students of all high schools who are interested in learning advanced physics.
- **Silver Medal** in Iran's National IMC. *2013*
This exam is annually held among students of all middle schools interested in mathematics.

Presentations and Invited Talks

- **OpenAirInterface workshop**: Tiny-twin: A CPU-Native Full-stack Digital Twin for 5G *2025*
Hosted by UT Austin, openairinterface
- **MobiCom OpenAI workshop**: Log Pruning for Efficient Q&A in 5G Networks *2025*
The 31st Annual International Conference on Mobile Computing and Networking Nov. 4-8, 2025, Hong Kong, China
- **6G Summit Abu Dhabi**: 5G Network Automation Using LLM Models and RAG *2024*
Hosted by IIT, UAE
- **Ericsson AI/ML/6G Weekly**: LLM for 5G: Network Management *2024*
Hosted by Amin Azari, Sweden
- **ICMLCN**: LLM for 5G: Network Management *2024*
IEEE International Conference on Machine Learning for Communication and Networking, Sweden

Selected Projects

Game Theory

- DA matching Algorithm Implementation* *Spring 2023*
○ Assigning patients with special disease types to hospitals using Python. Each hospital has a disease preference and each disease has a hospital preference.

Numerical Optimization

- Implementation of Optimization Algorithms* *Fall 2022*
○ Implementing line search based on strong Wolfe condition, steepest descent, Newton, and BFGS algorithms in MATLAB.

Massive Data Analysis

- City Traffic Analysis* *Fall 2022*
○ Finding hot spots and rush hour of a city based on camera data of cars in a city.

Deep Learning

Sentiment Detection

Fall 2022

- Fine-tuning multi-modal models and transformers to detect sentiment based on text and image.

Queuing Theory

Simulation of Q/Q/m queue

Fall 2022

- Simulating different Queuing scenarios including queues that contain positive and negative signals using Python.

Cloud Software Mobile Network

Routing Algorithm Implementation

Spring 2022

- Implementing dynamic routing with the Dijkstra algorithm on an SDN controller(ODL) in Python.

Advanced Data Networks

MIPS Processor Implementation

Spring 2022

- Implementing Multi-Cycle and Single Cycle architecture of MIPS Processor using Verilog.

Information Theory

Distributed Coding

Fall 2021

- Presenting and providing a term paper on distributed coding.

Data Networks

Simplified RPC Protocol

Fall 2021

- Implementing a simplified RPC protocol using socket programming and multi-threading in Python.

Machine Learning

Biometric Authentication

Spring 2021

- Biometric Authentication using ECG signals. Preparing noisy signals with MATLAB filtering toolbox and using SVM for classification.

Convex Optimization

Convex Optimization in Economics

Spring 2021

- Convex optimization on an economic problem using MATLAB and CVX.

Computer and Microprocessor architecture

MIPS Processor Implementation

Spring 2020

- Implemented Multi-Cycle and Single Cycle architecture of MIPS Processor using Verilog.

Programming Skills

MATLAB & Simulink

- Proficient in MATLAB programming (signal processing, CVX, digital filter design, and implementing machine learning algorithms and neural networks)
- Proficient in using Simulink to simulate communication systems and control systems.

Python Programming

- Proficient in programming using Python and experienced in working with libraries (CvxPy, sklearn, SciPy, TensorFlow, PyTorch, Spark, Pandas) and implementing learning algorithms using Python.

Other programming and tools: C/C++, go, bash, Kubernetes, ODL, NS3, GNS3, Git, Verilog, Assembly Mininet, Linux, Docker, Cuda

Selected Coursework

University of California San Diego

- Information Theory (A+)
- Modern Communication Networks(A+)
- Principles of Wireless Networks(A)
- Linear Algebra and Application(A)
- Computer Security(A+)
- Networked Services(A+)
- Parallel Computation(on going)

Sharif University of Technology

- Queuing Theory(20/20 - Graduate)
- Advanced Data Networks(19.5/20 - Graduate)
- Game Theory(20/20 - Graduate)
- Information Theory(19.1/20 - Graduate)
- Numerical Optimization(19.8/20 - Graduate)
- Data Networks(19.5/20 - Graduate)
- Random Processes(20/20- Graduate)
- Digital Communication(20/20)
- Convex Optimization(19.5/20)
- Communication Systems(19.6/20)
- Digital Signal Processing(20/20)
- Linear Control Systems(20/20)
- Introduction to Machine Learning(20/20)
- Probability and Statistics(19.7/20)

Language Skills

Persian: Native proficiency

English: Professional proficiency